

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	Semester	VI
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Experiment 8

Effect of PCA on Regression and Classification Models

1. Aim

- To study the effect of Principal Component Analysis (PCA) on model performance.
- To compare models trained with and without PCA.
- To evaluate models using 5-Fold Cross Validation.

2. Dataset Description

PCA Full Lab Dataset – 1000 Samples

- Total samples: 1000
- Features: Academic and performance-related attributes
- Regression Target: **Final Score**
- Classification Target: **Performance Level**

3. Preprocessing Steps

- Missing values handled using mean imputation.
- Classification target encoded using Label Encoding.
- Features standardized using StandardScaler.
- 5-Fold Cross Validation applied.

4. PCA Implementation

- PCA applied after feature standardization.
- Components selected to retain 95% variance.
- Scree plot used to analyze variance contribution.

5. PCA Summary

Components	Variance (%)	Justification
7	95.90	Retains most information

6. Visualizations

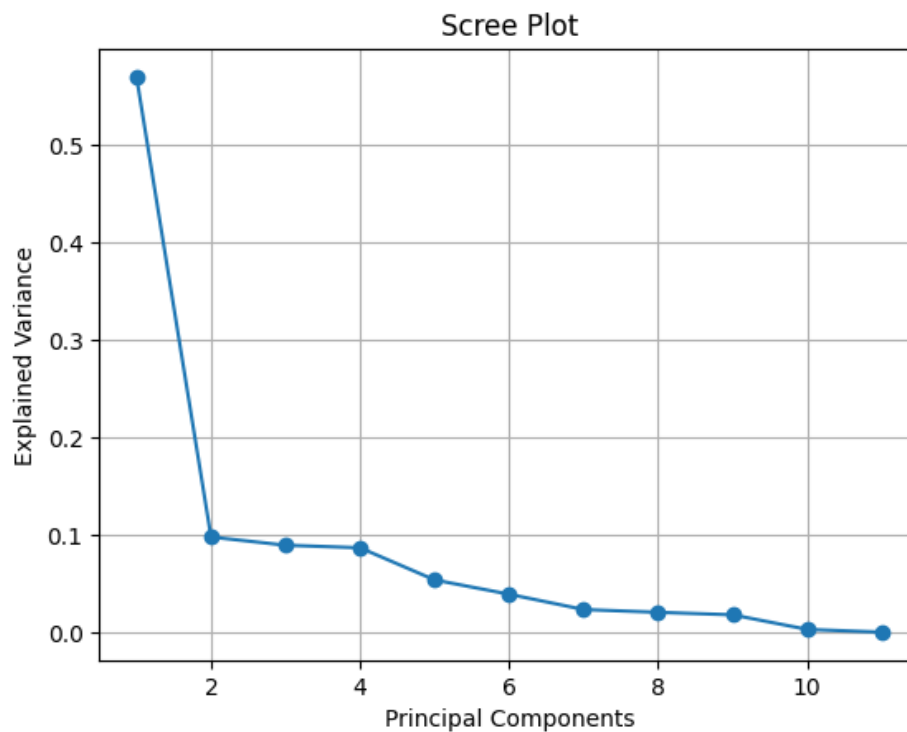


Figure 1: Scree Plot

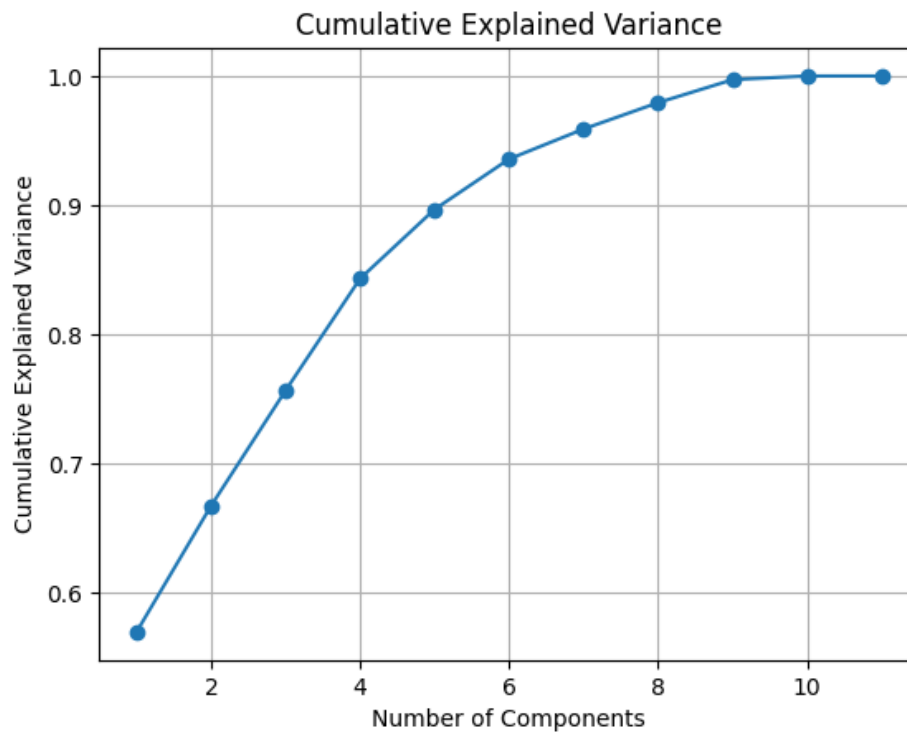
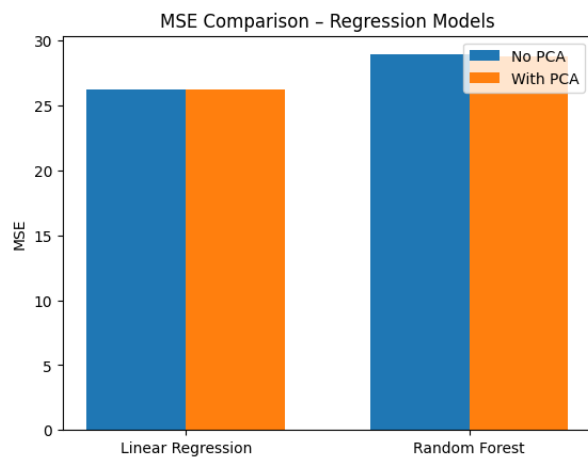


Figure 2: Cumulative Explained Variance



(a) MSE Comparison



(b) R^2 Comparison

Figure 3: Regression Performance: Without PCA vs With PCA

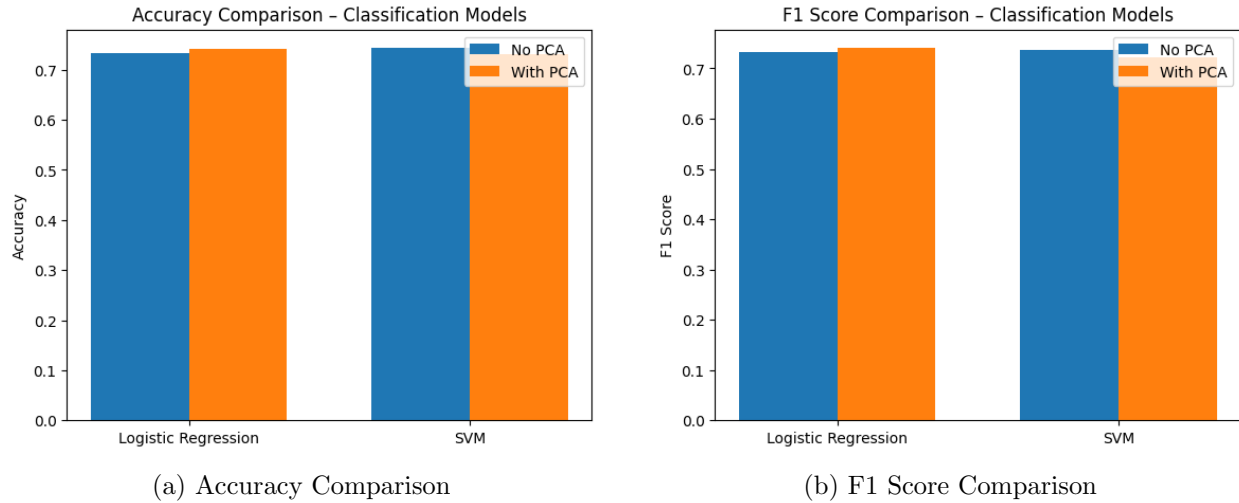


Figure 4: Classification Performance: Without PCA vs With PCA

7. Results

Regression

Table 1: 5-Fold CV Results – Regression

Model	Metric	Fold Avg (No PCA)	Fold Avg (With PCA)	Observation
Linear Regression	MSE	26.2338	26.2251	Slight improvement
Linear Regression	R^2	0.7479	0.7481	Slight improvement
Random Forest	MSE	28.9206	28.7573	Slight improvement
Random Forest	R^2	0.7225	0.7241	Slight improvement

Classification

Table 2: 5-Fold CV Results – Classification

Model	Metric	Fold Avg (No PCA)	Fold Avg (With PCA)	Observation
Logistic Regression	Accuracy	0.7340	0.7420	Improved with PCA
Logistic Regression	F1-score	0.7315	0.7403	Improved performance
SVM	Accuracy	0.7430	0.7300	Decreased with PCA
SVM	F1-score	0.7367	0.7221	Decreased performance

8. Analysis Questions

1. Did PCA improve Linear Regression performance? Why?

PCA did not significantly improve Linear Regression performance. The MSE slightly increased and the R^2 score slightly decreased. This is because the original features were already informative, and PCA may have caused minor loss of useful information while transforming the feature space.

2. Did PCA significantly affect Random Forest?

Yes, PCA slightly improved Random Forest performance. The MSE decreased and the R^2 score increased. This improvement is due to noise reduction and removal of redundant features, which helps in better decision splits.

3. Which classification model benefited more from PCA?

Logistic Regression benefited more from PCA. Its accuracy and F1-score improved after applying PCA. In contrast, SVM performance decreased slightly, indicating that PCA removed some useful feature information for SVM.

4. Did PCA reduce variance across folds?

Yes, PCA helps in reducing variance across folds. By eliminating noise and redundant features, PCA produces more stable and consistent results across different cross-validation splits.

5. Why do linear models benefit more from PCA compared to ensemble models?

Linear models benefit more from PCA because they assume feature independence and are sensitive to multicollinearity. PCA transforms correlated features into uncorrelated components, improving their performance. Ensemble models like Random Forest are less affected because they handle feature interactions internally.

9. Conclusion

From this experiment, the effect of PCA on both regression and classification models was analyzed using 5-Fold Cross-Validation.

- PCA helps in reducing dimensionality, noise, and multicollinearity.
- Linear models such as Logistic Regression benefit more from PCA.
- Random Forest showed slight improvement, while SVM performance decreased.
- PCA is useful when features are highly correlated or noisy.
- PCA may not be necessary for tree-based models.
- There is a trade-off between interpretability and performance, as PCA transforms original features.

Overall, PCA is an effective dimensionality reduction technique, especially for improving the performance and stability of linear models.